AETHERIA

Environmental Data Analytics and Weather Intelligence System

Course Project Report

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1. INTRODUCTION

Weather services provide large amounts of forecast data, which can be difficult to interpret directly. Aetheria addresses this challenge by converting hourly weather data into meaningful statistics, charts, and understandable insights. The application allows users to explore forecast conditions for different locations through a single dashboard.

2. PROBLEM STATEMENT

Raw weather data often requires additional processing to identify patterns, compare locations, and detect missing or suspicious records. Aetheria aims to provide a simple system that retrieves weather forecasts, analyzes the data, presents useful visualizations, and stores analysis summaries for future reference.

3. OBJECTIVES

- Retrieve weather forecasts for selected locations.
- Calculate descriptive statistics using Python and Pandas.
- Visualize temperature and precipitation-related information.
- Identify selected data-quality issues.
- Generate rule-based weather insights.
- Compare weather conditions across locations.
- Store analysis summaries in SQLite.
- Export data in CSV format through a user-friendly dashboard.

4. TECHNOLOGIES USED

- **Python:** Core programming and analysis.
- **Streamlit:** Dashboard and user interface.
- **Pandas and NumPy:** Data processing and numerical calculations.
- **Plotly and Matplotlib:** Data visualization.

- **Requests:** API communication.
- **Open-Meteo API:** Weather forecast and location data.
- **SQLite:** Persistent storage of analysis summaries.

5. METHODOLOGY

The system follows a sequence of data collection, processing, analysis, visualization, and storage.

Data Collection: The application retrieves location information and hourly weather forecasts through the Open-Meteo API.

Data Processing: The retrieved information is organized into Pandas DataFrames, and timestamps are converted into date-time format.

Data Analytics: The system calculates temperature statistics, humidity and wind summaries, precipitation information, and temperature changes across the forecast period.

Data-Quality Checks: The application examines missing values, duplicate records, suspicious temperature readings, and negative precipitation values.

Visualization and Insights: Charts display forecast patterns, while predefined rules generate simple explanations and an experimental weather risk indicator.

Storage and Export: SQLite stores analysis summaries, and CSV exports allow further examination of the data.

6. SYSTEM IMPLEMENTATION

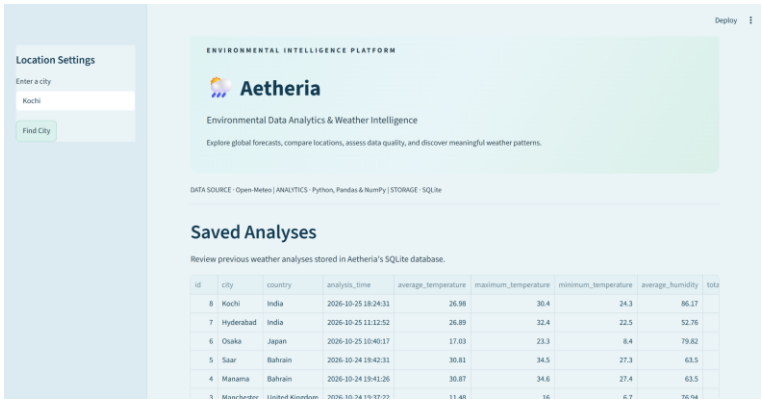
Aetheria is organized into separate Python modules for weather retrieval, analytics, data-quality checks, risk analysis, insights, and database operations.

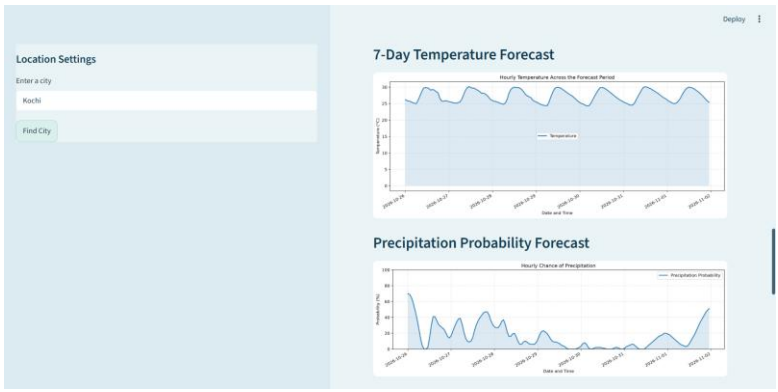
The Streamlit application connects these modules and displays weather metrics, forecast charts, analytical findings, and comparison results. Users can search for locations, analyze forecasts, compare selected cities, save analysis summaries, and export available data.

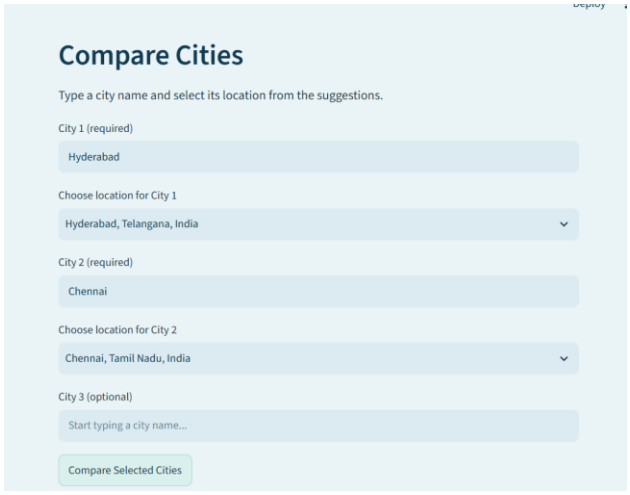
The SQLite database provides local persistent storage without requiring a separate database server. The modular design also makes the system easier to maintain and extend.

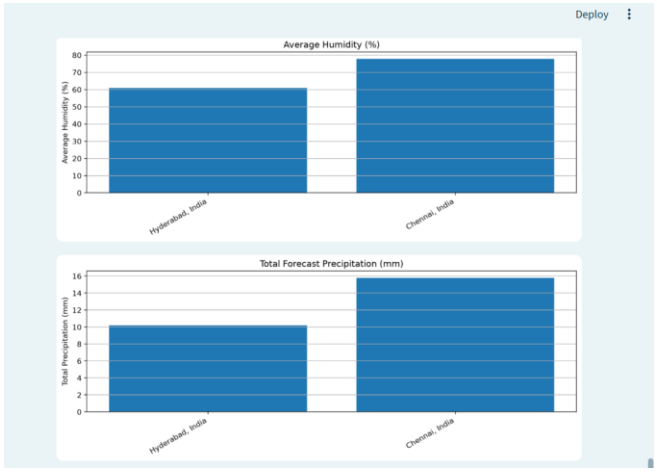
7. RESULTS AND DISCUSSION

The application integrates weather retrieval and analysis into a single dashboard. Its major outputs include statistical summaries, forecast visualizations, data-quality reports, rule-based insights, and comparisons between locations.


Deploy
Location Settings
Enter a city
Kochi
Find City
ENVIRONMENTAL INTELLIGENCE PLATFORM
Aetheria
Environmental Data Analytics & Weather Intelligence
Explore global forecasts, compare locations, assess data quality, and discover meaningful weather patterns.
DATA SOURCE · Open-Meteo | ANALYTICS · Python, Pandas & NumPy | STORAGE · SQLite
Saved Analyses
Review previous weather analyses stored in Aetheria's SQLite database.
id city country analysis_time average_temperature maximum_temperature minimum_temperature average_humidity tota
8 Kochi India 2026-10-25 18:24:31 26.98 30.4 24.3 86.17
7 Hyderabad India 2026-10-25 11:12:52 26.89 32.4 22.5 52.76
6 Osaka Japan 2026-10-25 10:40:17 17.03 23.3 8.4 79.82
5 Saar Bahrain 2026-10-24 19:42:31 30.81 34.5 27.3 63.5
4 Manama Bahrain 2026-10-24 19:41:26 30.87 34.6 27.4 63.5


Deploy
Location Settings
Enter a city
Kochi
Find City
7-Day Temperature Forecast
Hourly Temperature Across the Forecast Period
Temperature (°C)
Date and Time
Precipitation Probability Forecast
Hourly Chance of Precipitation
Precipitation Probability
Probability (%)
Date and Time


Compare Cities
Type a city name and select its location from the suggestions.
City 1 (required)
Hyderabad
Choose location for City 1
Hyderabad, Telangana, India
City 2 (required)
Chennai
Choose location for City 2
Chennai, Tamil Nadu, India
City 3 (optional)
Start typing a city name...
Compare Selected Cities


Deploy
Average Humidity (%)
Average Humidity (%)
Hyderabad, India
Chennai, India
Total Forecast Precipitation (mm)
Total Precipitation (mm)
Hyderabad, India
Chennai, India

The implementation demonstrates how Python libraries can transform raw forecast data into organized, visual, and understandable information. Database integration and CSV export provide additional ways to preserve and examine results.

8. LIMITATIONS

- Forecast accuracy depends on the external weather service and its models.
- Data-quality checks cover only predefined conditions.
- Insights and risk scores use simple rules and are not official weather warnings.
- Forecast comparisons do not establish long-term climate trends.
- The current implementation focuses primarily on forecast analysis.

9. FUTURE SCOPE

Future improvements could include historical climate analysis, additional environmental variables, expanded data validation, enhanced visualizations, and downloadable reports containing charts and statistical findings.

10. CONCLUSION

Aetheria demonstrates the practical application of Python data analytics, API integration, visualization, and database management. By combining statistical summaries, graphical representations, quality checks, and rule-based insights, the application makes weather forecast data easier to explore. The project provides a foundation for further development in environmental data analysis.

REFERENCES

1. Open-Meteo Weather API — https://open-meteo.com/
2. Python Documentation — https://docs.python.org/3/
3. Pandas Documentation — https://pandas.pydata.org/docs/
4. NumPy Documentation — https://numpy.org/doc/
5. Streamlit Documentation — https://docs.streamlit.io/
6. SQLite Documentation — https://www.sqlite.org/docs.html/